

the human brain. Of course, it wasn't like they were creating a shape like a brain and putting wires into it. Think of it more like a blueprint that formed the basis of the structure of early artificial intelligence devices. However, these quickly failed.

Dr. Asim Roy, a professor of Operations Research and Information Systems at Arizona State University, points out that these models are fundamentally flawed since they don't reproduce the external characteristics of the human brain, principally its independent way of learning. Therefore, methods based on these classical ideas require constant intervention by engineers and computer scientists – providing network designs, setting appropriate parameters correctly, and so on – to make them work. This drawback is severe, he maintains.

It was a thought that struck some other scientists early in the 1950s, and the attention of AI researchers started turning towards the psychological workings of the human brain rather than the physical workings.

Simulating how the brain thinks

It's in the importance of symbols that the working of the human brain and the history of artificial intelligence start intertwining. And the origin of it all couldn't be more innocuous. Some time in the 1950s, at the RAND corporation – a global non-profit think tank – a digital map was being printed. Those weren't the days of laserjets, so this humble printer was manipulating alphabets, numbers and punctuation marks to best simulate the map. Herbert Simon was witness to this ordinary act, which sparked an extraordinary thought process in his head.

While everyone around him saw a printer doing its job, Simon saw a machine using symbols – which is essentially what those alphabets, numbers and punctuations were – to convey a meaningful message in a medium it wasn't intended for. He was quick to realise that a machine that could manipulate symbols could potentially simulate the human thought process.

This realisation had a lot to do with Simon's own background as



Herbert Simon